

Exploring Socio-Economic Trends in England and Wales: A Visual Analytics Approach

Spring 2026

1. Overview

This coursework aims to give you experience of the whole lifecycle of carrying out a full visual analytics project. By completing this coursework, you will develop skills in exploratory data analysis, visual storytelling, and critical evaluation of data visualisations.

Your goals are:

- To follow a sound visual analytics process
- To develop a visualisation that displays important features of a dataset
- To write a clear report on your findings.

The outputs from this work should be

1. A Tableau dashboard and associate worksheets (**as a packaged workbook**: see https://help.tableau.com/current/pro/desktop/en-us/save_savework_packagedworkbooks.htm) - The dashboard should provide an interactive analysis of the selected data, highlighting key findings and allowing users to explore the dataset/findings. **If the dashboard is not submitted as a packaged workbook, the marker can't open it, and this will receive a zero mark!**
2. A written report with sections as defined below. The report should explain the background of the selected socio-economic issue, describe the data analysis process, and justify the chosen visualisation methods.
3. Python script

The submission deadline is **13:00 on Tuesday, 5th May** through Blackboard: create a single zip file containing all the files in your submission. This coursework is worth 100% of the marks for the unit.

2. Task Details

The task you are asked to carry out for the coursework is to design, construct, and evaluate an exploratory analysis of a complex dataset using both information visualisation *and* data projection. This dataset should be based on census data for England and Wales. You should design the visualisation to address some socio-economic issues that is important to you.

You must submit at least two data projections using different algorithms, implemented in Python using the methods covered in the labs. Each projection should result in a matrix representing the transformed data. Save these matrices in a file, such as CSV format, so they can be easily imported into Tableau for visualisation. Ensure that all files and scripts used in your work are saved and included in your submission. We also need to review the Python code used to generate the projections, so please provide it along with your data. The purpose of data projection is to reveal the data structure, including clusters, outliers, and relationships between different labels.

You will be using data from the last two censuses in England and Wales – 2011 and 2021.

The first set of data is taken from the 2011 census in England and Wales, which is indexed by the Excel file 2011CensusIndexofTablesandTopics_v11_4_2.xlsx. The tab labelled 'All Tables' provides a list of tables and links to the underlying data. You may find Tableau's Data Interpreter useful, and you may also need to edit some files to create usable datasets.

There are more than 1600 tables in total: clearly, this is far too many to create an interesting report. You should focus on a limited number of tables (probably around three or four) that

allow you to explore a particular aspect of socio-economic life in England and Wales: for example, health and links to nationality or occupation.

A new census was carried out in 2021 (during the pandemic). Some of the results have been released by the Office for National Statistics, but so far, these have only been in certain topics. A link to the topics that have been released can be found here <https://census.gov.uk/census-2021-results/phase-one-topic-summaries>. You should find that you can click through on a topic to a map display <https://www.ons.gov.uk/census/maps> and from here select a topic such as 'Housing'. Selecting a variable changes the map and also provides a link to download the data for that variable. Perhaps simpler is to visit the bulk downloads page https://www.nomisweb.co.uk/sources/census_2021_bulk

You need to use both the 2011 data and the 2021 data for at least one of your visualisations. Select 3-4 tables from the 2011 census and at least one dataset from the 2021 census for comparison. Use at least two different dimensionality reduction techniques (e.g., PCA, t-SNE) to visualise patterns or clusters in the data.

Something to note: Some geographic definitions don't necessarily match between the two census dates. This site will help you manage this

<https://www.ons.gov.uk/releases/censusmapsupdatechangeovertime>

You are expected to use Bayesian methods in any of the following ways:

- Census data often has missing values. Instead of using simple mean/mode imputation, students should apply Bayesian methods (e.g., Bayesian Linear Regression) to estimate missing values probabilistically.
- Use Bayesian models to infer missing socio-economic attributes (e.g., income levels in regions with sparse data).
- Apply Bayesian methods to predict trends in census data (e.g., housing affordability in 2030, employment trends).
- Use Bayesian posterior distributions to estimate credible intervals.
- Or any other way as long as you can justify Bayesian modelling choices.

Your report should contain the following sections:

- **Abstract.** A brief description of the key points in the report.
- **Introduction.** The background of the problem.
- **Data Preparation and Abstraction.** Describe the data manipulation necessary to create a dataset for analysis and the principal data types and semantics that you have analysed.
- **Task Definition.** A description of the tasks using Munzner's task taxonomy for which you have created the visualisations.
- **Visualisation Justification.** Define the visualisation techniques you use and justify your choices. You should refer to the principles of info vis, relevant aspects of human perception and cognition, and the scientific literature where appropriate. You should also explain why you have chosen the data projection methods that you have used. This justification and explanation are a very important assessment criterion, so do not skip on this and make sure that it is grounded in the theoretical concepts we have covered during the course.
- **Evaluation.** Using appropriate levels and types of validation (as in Chapter 4 of Munzner), assess the quality of your visualisation by making appropriate measurements and observations of the other students in your discussion group in an analytic task using your visualisation. (The list of discussion groups will be available on Blackboard).
- **Conclusion.** It is expected that you will address two aspects.
 - What you have learned about the socio-economic problem that was the basis of the visualisation.

- What you have learned about information visualisation from doing the coursework.

We are expecting the report to be about six to ten pages in length. This is an expectation, not a strict limit, so there will be no penalty for exceeding it. But if you find yourself writing much more than this, you are almost certainly providing too much detail. In particular, note that we will see the visualisation you generate, so there should be less or no need for screenshots.

We use the term 'dashboard' in the Tableau sense of a set of visualisations on a single screen. It is permissible to submit more than one Tableau dashboard or workbook if that supports the task better. Do not feel you have to squeeze everything onto a single dashboard.

Geocoding issues

It can be hard to plot the census data in Tableau because it does not contain outcome information. This [blog](#) contains some geocoding packages and a video on how to use them that support geographic information at many different levels of granularity. It should be helpful for you.

You may have some problems with using geocoding packages, in which case this link to Tableau help should be useful.

<https://kb.tableau.com/articles/issue/error-the-custom-geocoding-folder-has-errors-when-creating-map>

We have also provided a short guidance note on the Blackboard site, under assessment information.

3. Assessment

The assessment criteria are:

- **Problem understanding** (10 marks): How well you have explained the goals of the tasks, considering end-user requirements. Clearly define the socio-economic issue and its importance, identifying target users and their needs.
- **Data preparation and task analysis** (15 marks): How carefully you have extracted and manipulated the data, including insights gained through task analysis.
- **Data visualisation** (50 marks): How appropriate the visualisation and modelling approaches are, including systematic use of statistical and visualisation methods, with justification for the chosen approach. – 32 marks will be awarded for the visualisation, 10 marks for dimensionality reduction and 8 marks for justification.
- **Conclusions** (15 marks): How effectively you communicate what the user should learn from your analysis and what you have learned about large-scale data visualisation.
- **Presentation** (10 marks): How fluently and coherently the written text is structured, including the quality of images and graphics used. Writing should be clear, concise, and logically organised, with appropriate literature references.

Below are some general points that will help you when working on this coursework:

- Ensure that questions you set out to ask are answered by the visualisation and in the report.
- Having the option of switching between absolute values and proportions is often a useful feature. This is particularly helpful when comparing areas with different populations.
- When using dimensionality reduction, it is important to communicate to the user which variables were used in the original data space as otherwise, it is hard to interpret the plots.

- Tooltips should identify the corresponding point (e.g. a location), particularly for projected data.
- The introduction should contain some discussion of the type of user the visualisation is intended for.
- The report should note data anomalies (e.g. missing values) in the report, in particular, quantifying the number of missing values, etc.
- The abstract should describe the main findings of the work.
- Data cleaning matters.
- The use of section and page numbers helps the reader to navigate the report.
- References to secondary literature in your report's lit. review are valuable tools to provide context.

AI USE

This assessment is a **Category 1: prohibited (not allowed) use of AI**

You cannot use AI tools at all, including spelling and grammar checkers. For more information on AI use, please visit [Using artificial intelligence \(AI\) in assessments and for studying | Current students | University of Bristol](#)